

Comparing Fingerprint to Haptics-Based Authentication Methods in XR Devices

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ABSTRACT

Secure and seamless authentication in extended reality (XR) environments remains a significant yet understudied challenge. Traditional passwords are cumbersome with 3D controllers and novel methods are needed. We present a formative comparative evaluation of fingerprint to a haptics-based authentication methods for XR. Fingerprint authentication was implemented using an ESP-based capacitive sensor, while the haptics-based authentication was achieved through the replication of a Morse code-style vibration pattern displayed to the user. While fingerprint biometrics offer a fast login option, the haptics-based method does not require pre-registration, can be used as a one-time passcode, and does not require custom hardware.

1 INTRODUCTION

Immersive extended reality (XR) systems are no longer confined to entertainment and gaming. With the rapid growth of this field, XR increasingly supports sensitive applications such as collaborative research, education, healthcare, and cognitive science [2]. As these systems gain access to personal and private data, secure and XR-friendly authentication has become a critical design requirement [1]. However, authentication in XR remains challenging due to constrained input modalities, an increased risk of shoulder-surfing[4], limited visual attention, and the absence of standard text-entry interfaces. Traditional knowledge-based methods [6] such as PINs and passwords are error-prone and inefficient when entered using VR controllers, while biometric approaches often rely on dedicated sensing hardware that is not universally available across consumer XR devices. Prior research in human-computer interaction and virtual reality has explored physiological biometric authentication techniques, including fingerprint [5] and facial recognition, as well as alternative behavioral input modalities based on gestures, temporal patterns [7], keystroke dynamics and rhythm-based interaction. Among them, fingerprint authentication [5, 8] is well established and familiar among the mobile and desktop users due to its speed and low cognitive demand. However, its integration into XR systems typically requires external or headset-mounted sensors. In contrast, rhythm-based input methods, such as those using Morse code-style patterns [9] can be implemented entirely in software using standard controller input and haptic feedback, offering a potentially lightweight and hardware-independent authentication solution.

In this work, we present an empirical, within-subjects comparison between biometric and haptics-based authentication methods in XR. By contrasting a specialized hardware-assisted fingerprint method with a fully self-contained rhythmic, haptics-based input under controlled conditions, this study aims to provide early design insights into the trade-offs across efficiency, reliability, work-

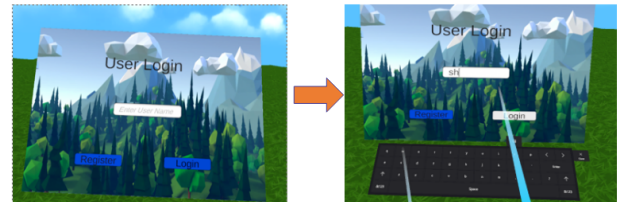


Figure 1: The initial interface for name registration and login with a pop-up virtual keyboard.

load, and usability for XR authentication workflows. Understanding these trade-offs is essential for informing the design of secure, accessible, and practical authentication mechanisms for current and future immersive systems. These findings aim to inform the design of secure, accessible, and practical authentication mechanisms for immersive systems.

2 SYSTEM ARCHITECTURE

2.1 System Components

The authentication prototype was developed in Unity using OpenXR. Figure 1 describes the user interface dynamically switches between fingerprint login and Morse code-style entry panels while sharing a unified component after registering the name and fingerprint samples. Figure 2 details the components of the authentication system. Core components include the TrialManager, which coordinates attempt scheduling and enforces practice constraints; the AuthController, which manages UI transitions and displays result feedback; and the RhythmController for Morse code-style trigger-press duration detection. Fingerprint authentication uses an ESP32-based microcontroller paired with a capacitive sensor to perform enrollment and matching through a lightweight Web-Socket. The VR controller trigger serves as the input mechanism for Morse code-style authentication, building on prior work [3].

2.2 System Layers

The system is organized into three interacting layers. The XR interaction layer handles task flow, user prompts, and input state transitions through TrialManager callbacks. The hardware interface layer integrates external components, forwarding verification requests to the ESP32 sensor during fingerprint mode and collecting trigger press timing data during Morse code-style mode. A communication and logging layer timestamps events, parses device messages, constructs authentication result records, and appends them to CSV logs for later statistical analysis.

3 EVALUATION

We conducted a comparative evaluation to understand tradeoffs between fingerprint and Morse code-style authentication methods.

Ten participants completed the study (4 female), ages ranging from 25 to 45 years, using a Meta Quest 2 head-mounted display (HMD). Authentication performance was analyzed across fingerprint and Morse code-style authentication methods, using both con-

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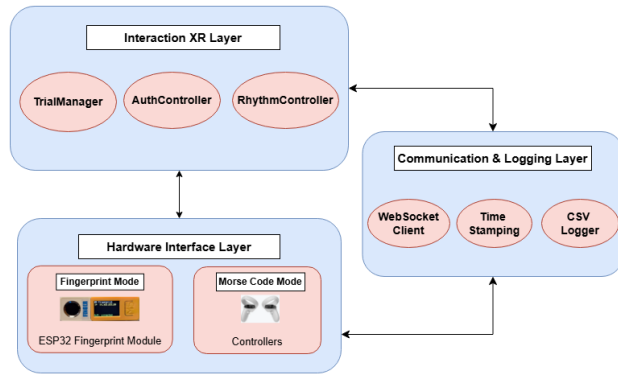


Figure 2: The system architecture of XR authentication using Fingerprint and Morse code-style input. The diagram illustrates core components and system layers.

tinuous (time) and binomial (success/failure) outcomes. Due to concerns regarding privacy, environmental robustness, and calibration overhead, iris-based authentication was excluded.

3.1 Procedure

During the study, each authentication method was attempted four times, beginning with a required practice trial followed by three data collection trials. The system prevented progression beyond practice until the attempt succeeded, ensuring participants' understanding of the tasks. In fingerprint mode, trial timing begins when login is initiated and ends when a success, mismatch, or timeout response is received from the ESP32, following best practices [2, 6]. In the Morse code-style method, timing begins at the first trigger press and ends after the full four-symbol code is entered and validated in four total phases. All trial-level outcomes, including timing, success or failure, and error codes, are recorded to maintain consistency between authentication techniques.

4 RESULTS AND DISCUSSION

A repeated-measures ANOVA revealed a substantial impact of authentication method on completion time, $F(1,9) = 338.22$, $p < .001$, with a very large effect size ($\eta^2 = .974$). Additionally, mean authentication time was noticeably faster for Fingerprint ($M = 3.80s$, $SD = 1.41$) compared to Morse code ($M = 39.92s$, $SD = 7.15$), demonstrating a dramatic performance advantage for Fingerprint authentication (Figure 3 left). For authentication accuracy, a Generalized Linear Mixed Model (GLMM) using binomial distribution revealed significantly higher success rates for Fingerprint relative to Morse code. Per-participant error rate (Figure 3 right) shows that Morse code authentication produced more failures, likely due to timing errors and symbol entry mismatches, whereas fingerprint failures were rare and attributed primarily to sensor-matching issues. In addition to performance, usability and workload ratings indicated a strong subjective preference for fingerprint authentication. NASA-TLX workload scores revealed substantially higher perceived physical demand, temporal demand, effort, and frustration during Morse code-style interaction. SUS scores further supported this pattern, with fingerprint authentication yielding higher usability ratings compared to Morse code, which aligns with the prior findings of biometric verifications [5, 7].

Although Morse code-style authentication demonstrated substantially slower performance and higher workload than fingerprint authentication, its unique characteristics suggest that it can be further investigated as a viable alternative. As user input relies solely on vibration-based feedback, it enables controller-based entry without requiring biometric sensors or physical contact, an advantage



Figure 3: Authentication time (left) and error rates (right) per participant for Fingerprint and Morse code method ($N=10$). $p < 0.01$, $\eta^2 = 0.932$ indicates a large effect size and error bars and individual data points show variation across participants.

for users wearing gloves, sharing devices, or operating in hygiene-sensitive contexts.

5 CONCLUSION AND FUTURE WORK

Across objective performance outcomes and subjective workload measures, fingerprint authentication consistently outperformed the alternative Morse code-style method, yielding substantially faster completion times, fewer errors, lower cognitive and temporal demand, and significantly higher usability ratings. Despite these performance limitations, Morse code remains appealing as a hardware-independent, privacy-preserving alternative that avoids biometric storage and can function when fingerprint sensing is unavailable or undesirable. Future work should investigate design improvements to haptics-based authentication methods to reduce overall input time as well as memory-based input errors through adaptive thresholding, calibration routines, or multimodal haptic/visual cues. Hybrid workflows that use a one-time method for initial authentication and biometrics-based continuous authentication is a promising direction for future research.

REFERENCES

- [1] M. Almaiah and T. Al-Hadhrani. Security and privacy issues in augmented reality: A survey. *Sensors*, 21(4):1136, 2021.
- [2] B. De Witte, V. Reynaert, D. Kieken, J. Jabbour, C. Demarey, A. Dumoulin, and J. Possik. Immersive virtual reality learning and cognitive load: A multiple-day field study. *Comput. Hum. Behav.*, 176, 2025.
- [3] J. G. Grandi, J. Terrell, K. Lofca, C. Ruizvalencia, and R. Kopper. A continuous authentication technique for XR utilizing time-based one time passwords, haptics, and kinetic activity. In *Proc. IEEE VR Abstracts and Workshops (VRW)*, 2023.
- [4] K. Krombholz, E. Busse, C. Pfeiffer, and M. Smith. Shoulder surfing defence for pin entry using haptic feedback. In *Proc. ACM CHI*, pages 1–10, Toronto, Canada, 2014.
- [5] D. Maltoni, D. Maio, A. K. Jain, and S. Prabhakar. Fingerprint recognition: A literature review. *IEEE Trans. Pattern Anal. Mach. Intell.*, 29(4):1–25, 2009.
- [6] S. Stephenson, B. Pal, S. Fan, E. Fernandes, Y. Zhao, and R. Chatterjee. SoK: Authentication in augmented and virtual reality. In *2022 IEEE SP*, pages 267–284, 2022.
- [7] H. Sun, W. Zhang, and L. Wang. Blink patterns as a biometric signal. In *International Conference on Biometrics*, pages 1–7, 2016.
- [8] G. Zheng et al. Finger-to-heart (f2h): Authentication for wireless implantable medical devices. *IEEE Journal of Biomedical and Health Informatics*, 23(4):1546–1557, July 2019.
- [9] X. Zhou, P. Li, and Q. Sun. Morse code-based authentication for secure manual input systems. *IEEE Access*, 8:140120–140132, 2020.